Q–110

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BCA II Sem Main/Back Examination 2025-26

BCA–201—Mathematical Fundamentals

Time : 3 Hours] [Max. Marks : 70

[Min. Passing Marks :

Instructions to Candidates :

Part–A : Short Answer Type Questions (up to 25 words) 10 × 2 = 20 marks. All 10 questions are compulsory.

Part–B : Analytical/Problem Solving questions 5 × 4 = 20 marks. Candidates have to answer 5 questions out of 7.

Part–C : Descriptive/Analytical/Problem Solving questions 3 × 10 marks = 30 marks. Candidates have to answer 3 questions out of 5.

Schematic diagrams must be shown wherever necessary. Any data you feel missing may suitably be assumed and stated clearly. Units of quantities used/calculated must be stated clearly.

Use of following supporting materials is permitted during examination. (Mentioned in form no. 205).

1 _____ 2 _____

Part–A

1. Define a Proposition with example ?
2. What is an argument in logic ?
3. What is a Venn diagram ? Explain its uses.
4. Define :
   (i) One-one (injective) function
   (ii) Onto (surjective) function
5. Define distributive property.
6. Define Boolean algebra.
7. Given $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$, $B = \begin{bmatrix} 2 & 0 \\ 1 & 5 \end{bmatrix}$, then find :
   (i) A + B
   (ii) A – B
8. Find the adjoint of the matrix : $A = \begin{bmatrix} 2 & 1 \\ 5 & 3 \end{bmatrix}$
9. Explain different methods of data collection.
10. Define standard deviation and variance ?

Part–B [5×4 = 20]

1. Construct the truth table and determine whether the following statement is a tautology :
   $(p \rightarrow q) \rightarrow (q \rightarrow p) \rightarrow (p \rightarrow r) \rightarrow r$
2. Let relation on integers is de lined by :
   a – b if a – b is divisible by 2
   Check whether it is an equivalence relation.

Q. 3. Solve using Inclusion-Exclusion Principle : In a survey of 120 students : 70 like Cricket, 60 like Football, 40 like both then find :

(i) Students who like at least one

(ii) Students who like neither

Find out the inverse of matrix

Q. 4. Find out the inverse of matrix $\begin{bmatrix} 1 & -1 & 2 \\ 4 & 0 & 6 \\ 0 & 1 & -1 \end{bmatrix}$ (2)

Q. 5. Solve the following system using matrix method :

$$3x + 2y = 7$$
$$2x - y = 1 \qquad (4)$$

Q. 6. Discuss classification and tabulation of data with examples.

Q. 7. Explain types of correlation with examples.

Part-C **[3×10 = 30]**

Q. 1. (a) Determine whether the following formulae are tautology, contradiction or satisfiable :

(i) $\{P \rightarrow (Q \rightarrow (P \wedge Q))\} \leftrightarrow P \rightarrow +$

(ii) $[(P \rightarrow Q) \wedge (Q \rightarrow R)] \wedge (P \wedge \sim R)$ (5)

(b) Explain Proposition and Logical Operators in detail with examples.

Q. 2. (a) Let the universal set U = {1, 2, 3, 4, 5, 6, 7, 9, 10}, Let A = {1, 2, 3, 4, 5}, B = {2, 4, 6, 8} and C = {1, 4, 7, 10}, then find the following :

(i) $A - (B \cup C)$

(ii) $(A \cup B) - (C - B)$

(iii) $B^c - (C - A)$

(iv) $(A^c \cup B) \cap (C^c - A)$ (5)

(b) If $f(x) = 2x^2 + 1$, $g(x) = x^2$ then find

(i) $(fog)(x)$

(ii) $(gof)(x)$ (5)

Q. 3. (a) How many 4-digit numbers can be formed using digits 1–7 :

(i) Without repetition

(ii) With repetition

(iii) Divisible by 2 (5)

(b) A group consists of 7 men and 5 women. In how many ways can a committee of 4 persons be formed such that :

(i) At least 2 women are included ?

(ii) At most 3 men are included ? (5)

Q. 4. Solve using Cramer's Rule :

$$x + y + z = 6$$
$$2x + 3y + z = 10 \qquad (5)$$
$$x + 2y + 3z = 13$$

Q. 5. Find Karl Pearson's coefficient of correlation for the data :

X	Y
1	2
2	4
3	5
4	4
5	5

(5)

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